RIPHAH
INTERNATIONAL UNIVERSITY

BS-AI

LAB TASK 02

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Sap I'D: 72405

Subject: Object Orientated Programming

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Task 1:

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Name: Hafiz Abdullah Shakeel\n";
    cout << "SAP ID: 72405\n\n";

    int rows, cols;
    cout << "Enter number of rows: ";
    cin >> rows;
    cout << "Enter number of columns: ";
    cin >> cols;

    int matrix[100][100];

    cout << "Enter elements of the matrix:\n";
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            cin >> matrix[i][j];
        }
    }

    int primeCount = 0;
    cout << "\nPrime numbers in the matrix:\n";
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            int num = matrix[i][j];
            bool prime = true;
```

```
            if (num <= 1) prime = false;
            else {
                for (int k = 2; k * k <= num; k++) {
                    if (num % k == 0) {
                        prime = false;
                        break;
                    }
                }
            }

            if (prime) {
                cout << "Element at (" << i << "," << j << ") = "
                     << num << " is prime.\n";
                primeCount++;
            }
        }
    }

    cout << "\nTotal prime numbers in the matrix: " << primeCount << endl;

    return 0;
}
```

Task 2:

```
#include <iostream>
using namespace std;

int main() {
    cout << "Name: Hafiz Abdullah Shakeel\n";
    cout << "SAP ID: 72405\n\n";

    int rows, cols;
    cout << "Enter number of rows: ";
    cin >> rows;
    cout << "Enter number of columns: ";
    cin >> cols;

    int matrix[100][100];

    cout << "Enter elements of the matrix:\n";
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            cin >> matrix[i][j];
        }
    }
```

```
int oddCount = 0;
cout << "\nOdd numbers in the matrix:\n";
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < cols; j++) {
        if (matrix[i][j] % 2 != 0) {
            cout << "Element at (" << i << "," << j << ") = "
                << matrix[i][j] << " is odd.\n";
            oddCount++;
        }
    }
}

cout << "\nTotal odd numbers in the matrix: " << oddCount << endl;

return 0;
```

Task 3:

```
MyJavaFirstProgram.java x   Cal.java x
rce   History
package myjavafirstprogram;
import java.util.Scanner;
import javafirstclassbsai.Cal;
public class MyJavaFirstProgram {
        public static void main(String[] args)
        {
            Scanner in=new Scanner (System.in);
            Cal c=new Cal ();
            System.out.println("Enter values");
            int a,b,result=5,mul;
            a=in.nextInt();
            b=in.nextInt();
            //c.sum(a,b);
            mul=a*b;
            System.out.println("The sum is" +result+" Multiplication is "+mul)
        }

        }
```

```java
package javafirstclassbsai;
public class Cal {
    void sum (int a, int b){
        int res=a+b;
        System.out.println(res);
    }
}
```

put - Java First Class BSAI (run)

```
run:
Enter values
6
9
The sum is5 Multiplication is 54
BUILD SUCCESSFUL (total time: 6 seconds)
```